Orthopolity: a general law of resource allocation

Capacity, physical realizations, and effective-resource inference

Research manuscript, revised 14 October 2026. Detailed derivations, complete comparisons and analysis histories are retained in the supporting material. Empirical studies include exploration, locally frozen forecasts and retrospective reanalyses; none is externally preregistered. Authorship and submission declarations require agreement before submission.

Abstract

Orthopolity states a general law of resource allocation: Nature tends to distribute resources equally among concentrations of those resources. We develop its mathematical content, establish exact realizations under specified physical conditions, and examine its predictive reach in models and data. Finite resource capacity bounds incidence; complete partitions give an exact inverse relation between count and mean resource. Allocation among coexisting classes requires additional conditions, including an independently defined comparison measure. Symmetric resource exchange supplies one mechanism, while gravity, radiating waves and thermal modes clarify which geometric, transport or statistical equality is present. Absorption, quantization and cell-cycle variation supply independently specified departures. The framework also supports inverse inference: restricted combinations of measured resource requirements produce identifiable predictions across environments. Tests show accurate allocation forecasts in an engineered system, pure-to-mixture chemical competition, approximate budget closure across separately grown algal cultures, and predictive gains from combined symbolic costs in language. A conditional redistribution bound shows why changing resource prices alone cannot explain one biological failure. Other forecasts fail, and some natural catalogues cannot support the intended test after observation and estimator checks. These outcomes distinguish an established conditional relation, a physically explained departure and an unresolved failure. The contribution is a common account of resource incidence and allocation that makes those distinctions operational and connects them to further predictions. Establishing its general physical reach requires both successful transfers and explicit boundaries; failure is retained as information about the model and its domain.

1. The law and the scientific task

Nature tends to distribute resources equally among concentrations of those resources. This is the general-law statement of orthopolity. The article seeks to establish its physical content through exact ideal systems, realizations in known laws, mechanisms for real systems, and empirical predictions. It follows the accounts of Fabbri and Oliveira Jr. (2017) and Fabbri (2024), while distinguishing the claims that their shared intuition can support.

A box of volume B accommodates B/q equal smaller boxes of volume q when they tile it completely. At every complete level of a nested subdivision, more units mean less resource per unit and the same total resource. A mixed population presents an additional question: how much of the total is assigned to each coexisting size class? Conservation answers the first accounting question but does not settle the second allocation question. Their connection, and the conditions needed to make it, are central to the law's application.

An equal allocation can appear unequal when expressed per a different coordinate, when the relevant resource combines several requirements, or when another process changes its distribution. Equal power through enclosing surfaces accompanies inverse-square radiative flux density. Equal classical energy per mode accompanies a spectrum increasing with mode density. These examples require explicit objects, resources and measures; an exponent by itself does not identify the equality. Ecological size-spectrum work provides an important precedent for making these choices physical (Sheldon et al., 1972; Gaedke, 1993; Hatton et al., 2021).

The law's scientific value also concerns departures. A standing tree experiences gravity while other forces support it; the absence of downward motion does not identify which forces act. Likewise, unequal observed allocation can coexist with an equalizing process. Establishing that explanation requires independent information about the processes and a prediction of their combined effect. A fitted residual called a constraint supplies neither. Some departures will instead expose a failed resource choice, an inapplicable mechanism or a limitation of the allocation relation in the claimed domain.

The paper therefore develops three connected contributions: a precise account of capacity and allocation; a way to infer restricted effective resources and predict across conditions; and comparisons that preserve both successful forecasts and failures. Additivity, reversible dynamics, equipartition and the statistical tools have established foundations. The contribution concerns their physical organization, the consequences that can be transferred between settings, and the limits revealed by testing them. Supporting material retains the detailed proofs, secondary examples, complete comparisons and source qualifications.

2. Capacity, allocation and inference

2.1 An exact resource-incidence core

Let ν be a nonnegative additive resource measure on a system Ω, with finite budget $B = \nu(\Omega) > 0$. For disjoint units A_i with requirements $q_i = \nu(A_i)$, finite additivity gives

$$N_{\geq t} = \#\{i : q_i \geq t\} \leq B/t, \qquad t > 0.$$

Indeed, $tN_{\geq t} \leq \sum_{q_i \geq t} q_i \leq B$. This is a capacity bound; attaining it requires feasibility and use of the relevant budget.

For a nonempty finite description at level j, define its count N_j, mean resource $\bar{q}_j$, and multiplicity $m_j(x) = \sum_i \mathbf{1}_{A_{ji}}(x)$. Integration gives

$$N_j \bar{q}_j = \int_\Omega m_j \, d\nu = B\gamma_j, \qquad \gamma_j = \frac{1}{B} \int_\Omega m_j \, d\nu.$$

Every complete partition has $\gamma_j = 1$ and therefore $N_j \bar{q}_j = B$. The inverse count–mean-resource relation holds exactly even for unequal units and nested levels. Incomplete disjoint coverage has $\gamma_j < 1$; overlap can give $\gamma_j > 1$. Independently measured coverage consequently predicts a correction to the inverse relation.

A complete level accounts for the whole resource once. Several levels describe that same budget repeatedly; their totals are not independent allocations. For coexisting disjoint classes, fractions f_j instead give $N_j \bar{q}_j = B f_j$ with $\sum_j f_j \leq 1$. Equal class allocations require equal fractions, which is an additional physical statement. These elementary results establish the exact core and identify where an allocation law adds content (supporting derivation).

2.2 The allocation relation and its measure

Let k denote size on a declared finite domain and let μ be the reference measure of the classes being compared. If $n_\mu = dN/d\mu$ and $\bar{q}(k) = E[q \mid k]$ is the arithmetic mean additive resource per object, then resource density obeys

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k) n_\mu(k). \tag{1}$$

Neutral orthopolic allocation relative to μ has

$$\mathcal{O}_\mu(k) = C > 0, \qquad n_\mu(k) = \frac{C}{\bar{q}(k)}. \tag{2}$$

This concerns an allocation density, an expectation or a time average as specified by the system. Those interpretations are not interchangeable. Resource-weighted sampling gives a uniform size distribution under (2), while sampling objects uniformly gives an inverse-cost distribution. Stocks, accumulated requirements and throughputs must also be distinguished before applying the relation.

The reference measure is physical input. If $\bar{q}(k) \propto k^d$, equal resource per $d \ln k$ gives the **log-orthopolic** count density $dN/dk \propto k^{-(d+1)}$. Equal resource per dk instead gives the **linear-orthopolic** density $dN/dk \propto k^{-d}$. A change of coordinate carries a Jacobian and need not preserve a visually flat plot. Choosing the measure after seeing an abundance profile can manufacture equality, so an empirical claim must justify that choice independently.

One sufficient allocation condition is exchangeability of resource shares among K fixed equally weighted classes, with fixed total R. Symmetry gives $E[R_i] = R/K$ and, for fixed positive cost q_i, $E[N_i] = R/(Kq_i)$. Conservation alone does not imply this symmetry. Nor does equal expectation imply equal snapshots: placing the entire resource in one randomly chosen class has symmetric expectations and unequal realizations. The observation model must represent this distinction.

2.3 Effective resources, constraints and transferable predictions

Objects can require several resources. With independently fixed reference quantities s_ℓ, write

$$Q_i = Q_0 F(q_{i1}/s_1, \ldots, q_{im}/s_m; z_i, \theta), \qquad \bar{q}_{\text{eff}}(k) = E[Q_i \mid k].$$

The total $\sum_i Q_i$ is additive over the declared object partition, but a nonlinear F need not preserve that total when objects are split or merged. Proportional subdivision requires degree-one homogeneity for invariance. If all nonnegative constituent vectors and their subdivisions are admissible, invariance under every heterogeneous subdivision forces a finite nonnegative F, with $F(0) = 0$, to be a nonnegative linear form. Nonlinear effective resources therefore need a specified physical partition or restricted composition domain. With heterogeneous objects, averaging follows composition: $E[F(X, Y)]$ generally differs from $F(E[X], E[Y])$.

Several constituent budgets also remain several constraints unless a physical model justifies reducing them. Feasibility can require $\sum_i N_i q_{\ell i} \le B_\ell$ for each constituent. A specified optimization problem may provide scarcity-weighted effective costs, while a separately supported identical population has capacity $\min_\ell B_\ell/q_\ell$. Neither scalarization nor capacity alone fixes allocation among competing types (Boyd and Vandenberghe, 2004).

Write a constrained allocation as

$$\bar{q}_{\text{eff}}(k) n_\mu(k) = Ca(k), \qquad a(k) > 0. \tag{3}$$

A mechanism or independent measurement must determine a for a forward prediction. Inversion identifies $\bar{q}_{\mathrm{eff}}/a = C/n_\mu$, up to scale. Thus abundance contains information about an effective resource, but cannot by itself separate that resource from an unrestricted constraint. Multiplying both by the same positive function leaves the prediction unchanged.

Restrictions shared across environments make the inverse use predictive. Consider the composition $Q = Q_0 \prod_{\ell=1}^{m} X_\ell^{\theta_\ell}$. Suppose independently measured constituent requirements scale as $X_{s\ell} \propto k^{D_{s\ell}}$, and a known allocation factor is $a_s \propto k^{-v_s}$. Under logarithmic allocation, a count density $k^{-(1+\alpha_s)}$ implies

$$b = D\theta, \qquad b_s = \alpha_s - v_s. \tag{4}$$

This requires deterministic class requirements, or size-independent residual moments after combining heterogeneous constituents. For unrestricted coefficients, full column rank identifies the shared composition; otherwise only some combinations may be identified. A new constituent row has a unique forecast when it lies in the row space of the calibration design. Every left-null vector gives a parameter-free consequence $\ell^\top b = 0$.

For example, calibration rows $(1, 1)$ and $(1, 2)$ identify a two-resource composition. A new row $(2, 1)$ then predicts $b_3 = 3b_1 - b_2$ before the third abundance slope is measured. Uncertainty and near-collinear constituent scalings determine whether that forecast is useful. The identification study verifies these standard inverse-problem calculations synthetically; it does not count constructed data as natural evidence. Inverse optimization and inverse statistical mechanics provide related precedents (Keshavarz et al., 2011; Habeck, 2014). Here the scientific content lies in the specified resource family and its transferred physical predictions.

3. Ideal mechanisms and physical realizations

3.1 Equalization, a constraint and its removal

Consider fixed classes with positive reference widths w_i, fixed costs q_i, and resource stocks R_i totaling B. Symmetric conductances $g_{ij} = g_{ji} \geq 0$ on a connected graph give the exchange current

$$J_{ij}^0 = g_{ij} \left(\frac{R_i}{w_i} - \frac{R_j}{w_j} \right).$$

The resulting dynamics conserve resource and relax to $R_i^0 = B w_i / \sum_j w_j$, hence $N_i^0 = R_i^0/q_i$. This mechanism selects equal resource per the declared measure. It establishes the neutral relation within its specified model class.

A fixed class preference V_i changes the transition rate from i to j to $(g_{ij}/w_i)\exp[\max(V_i - V_j, 0)]$. Detailed balance then gives

$$R_i^V = \frac{B w_i e^{-V_i}}{\sum_j w_j e^{-V_j}}.$$

The added rates balance the neutral current at constrained equilibrium. Removing the preference preserves the current state and predicts relaxation toward the neutral allocation. The generator's spectral gap supplies a convergence bound; disconnected components equalize only within their separately conserved budgets. These are established reversible-chain tools applied to resource classes (Levin and Peres, 2017).

This construction shows how an active equalizing process can have an unequal outcome. Identifying that decomposition in a natural system requires evidence about its kinetics or response to intervention; a stationary profile alone is insufficient.

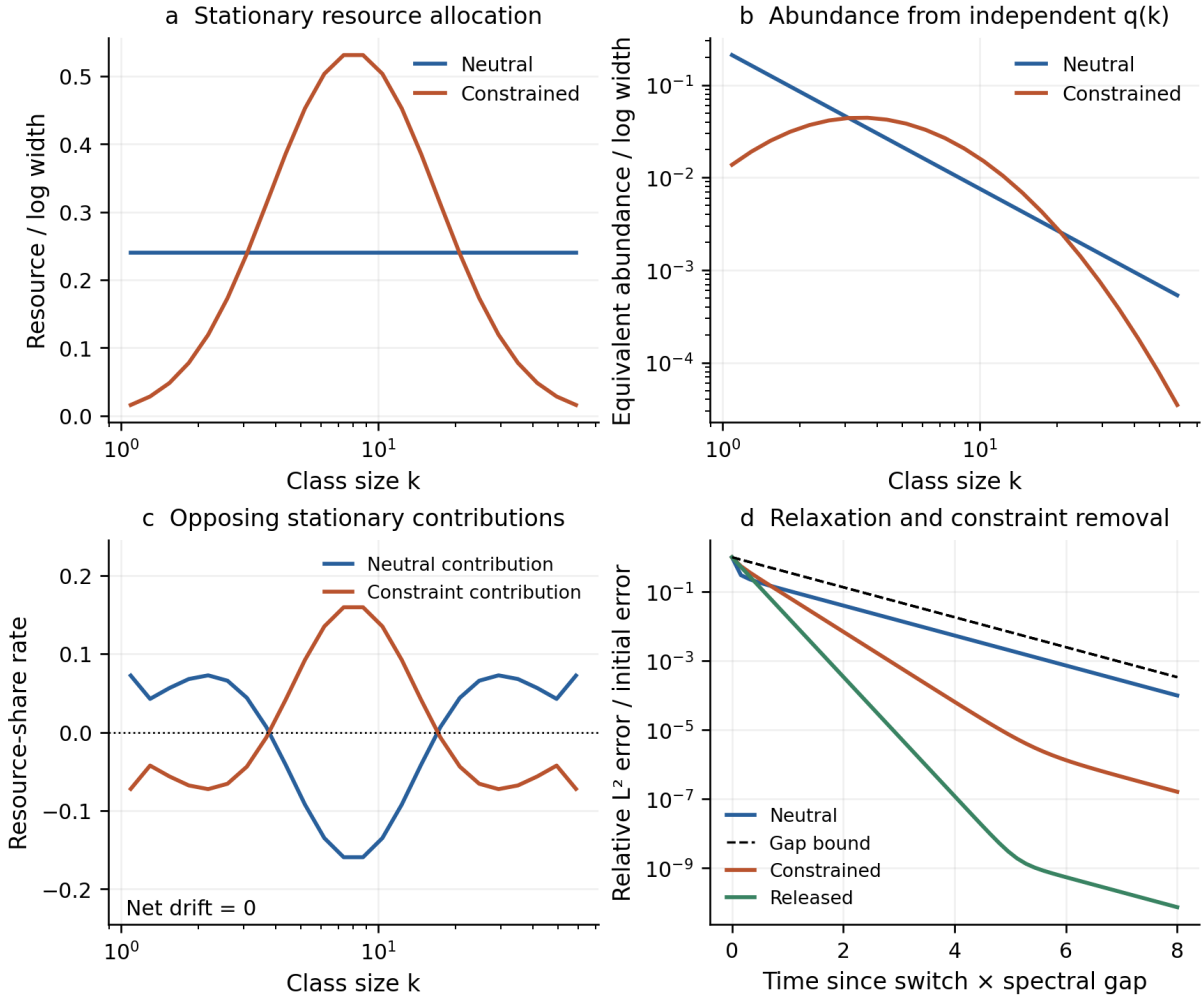


Figure 1. Resource exchange on fixed logarithmic size classes. The model predicts neutral allocation, a nonuniform constrained equilibrium and recovery after release. The detailed model, spectral bounds and finite-packet simulations are in the supporting material and model report.

3.2 Gravity and radiating waves: what geometry preserves

For a Newtonian gravitational field $\mathbf{g}$, the Poisson equation (Chao, 2014) gives the integral relation $\oint_S \mathbf{g} \cdot \mathbf{n}\, dA = -4\pi G M_{\text{enc}}$. Every enclosing surface outside a fixed isolated source has the same inward field flux. Spherical symmetry additionally gives $|\mathbf{g}| = GM/r^2$: the conserved enclosed-source quantity is distributed over an area growing as r^2. Inward flux per solid angle is $d\Psi/d\Omega = GM$, an explicit angular equality. This is a geometric realization with specified symmetry. The field flux is not an energy throughput, and nested surfaces are not independent mass or energy shares. Distributed sources change $M_{\text{enc}}(r)$; nonspherical sources need not give uniform local fields. The gravity derivation states the resource interpretation, energy-accounting limits and departures.

For radiation, Poynting's balance gives a physical energy throughput. In steady transparent transport outside a source, the enclosing-surface power is $P(r) = L$; isotropic emission gives flux density $L/(4\pi r^2)$. With outgoing radial propagation at speed c, disjoint shell stocks instead

satisfy $dU = (L/c)dr$. Equal radial thicknesses hold equal energy. Equal logarithmic radial intervals have $dU/d\ln r = Lr/c$, so they do not have equal stocks. The geometry identifies which comparison realizes equality.

An independently specified absorbing medium gives

$$P(r) = L \exp\left[-\int_{r_0}^{r} \kappa(s)\, ds\right].$$

It predicts the departure from constant radiative throughput and accounts for energy deposited in matter. If absorption is removed while the source continues, the response travels outward and the transparent profile returns by $(r - r_0)/c$. The transport derivation gives the full space-time response and its assumptions (Poynting, 1884; Haus and Melcher, 1989; Condon and Ransom, 2016). Neither the gravitational nor radiative inverse-square law alone determines the abundance of objects at different sizes.

3.3 Thermal modes: a derived limitation of neutral allocation

A finite set of classical harmonic modes in canonical equilibrium has mean energy $k_B T$ per mode. Counting modes is the independently defined measure; instantaneous mode energies can fluctuate. For electromagnetic excitation energy above the ground state, quantization gives instead

$$\bar{\epsilon}_\nu = \frac{h\nu}{e^{h\nu/(k_B T)} - 1} = k_B T \frac{x}{e^x - 1}, \qquad x = \frac{h\nu}{k_B T}.$$

The classical plateau is approached for $x \ll 1$ and suppressed for $x \gg 1$. The departure follows from the allowed states before a spectrum is observed. Mode density $8\pi\nu^2/c^3$ then gives Planck's spectrum. Equal energy per mode is therefore different from equal energy per frequency interval (Planck, 1901; Tong, n.d.).

Our FIRAS calculation expresses the published microwave background spectrum as excitation energy per mode. Its 43 channels span $x = 1.199$–11.262, illustrating quantum suppression rather than the classical plateau. The distributed product already contains a fitted blackbody template; this calculation preserves established observational evidence and its correlated errors, without claiming a new independent confirmation (Fixsen et al., 1996). The full derivation and spectrum are retained in the supporting material.

A nonequilibrium test asks more: can independently fitted damping predict mode energy under a temperature gradient? Using Fontana and Bellon's (2023) cantilever archive, we fixed a quadratic local-loss model on colder conditions and sealed hot-condition forecasts before evaluating fluctuations. Ten of eleven mode fits have nonnegative inferred local loss. On their 26 targets, mean absolute error is 74.0 K, versus 78.9 K for uniform local loss on the same modes. Individual transfers range from close agreement to large errors; one fit requires negative local dissipation. Thus measured global damping does not by itself identify an adequate spatial energy weighting. The retrospective design, calibration conventions, all failures and a source point hidden by the plotting range are reported in supporting section S5.13. Equilibrium normalization was inherited, so this is not a new test of equipartition.

3.4 Cell growth and division: a measurable constraint envelope

Let additive cell mass x grow as $\dot{x} = gx$, with common deterministic g, conservative symmetric binary division and no death. In a balanced asynchronous population with a stationary lineage of physiologically equivalent daughters, established growth-fragmentation theory gives $p_L(x) = xp_P(x)/E_P[x]$ (Genthon, 2022). For lineage birth mass B and division mass D, stationarity requires $D \overset{d}{=} 2B$. The fraction of cycles crossing size x is

$$H(x) = P(B \le x < D) = F_B(x) - F_B(x/2),$$

and the population law becomes

$$p_P(x) = \frac{2H(x)}{E[B^{-1}]x^2}, \qquad \frac{x^2 p_P(x)}{E_P[x]} = \frac{H(x)}{\ln 2}.$$

The right-hand side is the normalized mass allocation per logarithmic size. Where every cycle crosses an interval, $H = 1$ and allocation is log-orthopolic. Fixed birth mass b gives the inverse-square density $2b/x^2$ on $[b, 2b)$; variable birth masses can remove any exact plateau. Independently measured birth events predict that envelope. Growth noise, asymmetric division or physiological lineage selection require a different observation model. This provides both a realizable ideal relation and a specified route to its failure under changed assumptions (derivation).

4. What the empirical comparisons establish

4.1 Predictions and comparisons

A test specifies its objects, measure, resource or cost, constraints and observation model, then evaluates a forecast on additional observations. Prediction after independent calibration is distinct from an identity evaluated on the same data. Our studies retain frozen inputs, forecasts, comparators and outcomes in a run registry. Published summaries were sometimes known and some raw outcomes had been exposed; the supporting material records that chronology. Freezing a later analysis does not erase earlier exposure.

Evidence	**Established outcome**	**Scope or unresolved limitation**
Runtime resource competition	Calibrated costs predict allocation and a thread-count intervention	One engineered mechanism
Size-selected algal cultures	Equal-biovolume forecasts transfer across selection treatments	Separate populations; binding resource unmeasured
Linguistic genre transfer	Combined letter/phoneme cost improves over letter-only models	Lexical frequencies predict better; physical cost unidentified
Bacterial summary transfer	Population mean predictions are close	Variability and the joint comparison fail
Cantilever energy transfer	Some damping-to-energy forecasts transfer	Small average gain; spatial-loss and observation failures remain
Chemical / sediment constraints	Independent inputs improve transfer forecasts	Established rivals perform similarly or better; finite-accuracy limits
Chemostat nutrient pulse	Frozen two-budget reweighting fails	Constraint response needs explanation
Flare and lake geometry	Predicted exponents excluded or equivalence unresolved	Measurement and dependence affect interpretation
Seismicity	Qualification or estimator prerequisites fail	No real b–dimension comparison established

Table 1. Different studies answer different questions. Detailed results, all rival forecasts and observation limitations are retained in supporting material.

4.2 An allocation mechanism with an intervention

In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes compete for one execution resource. Independent calibration gives cost degrees 1.870 and 2.686 for two kernels. The complete mean-cost curves predict CPU and job profiles before 16 allocation trials: maximum CPU-share errors are 0.0035–0.0063 and job-share total-variation errors are 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produces the predicted allocation change within 0.0078. A unit-cost alternative has job-share errors of 0.126–0.273 (intervention report).

The system supplies competition and the intervention changes the independently counted opportunities. This is stronger evidence about allocation than assigning quotas and checking their cost. It demonstrates one physical implementation of a conditional resource law; it does not establish the same mechanism in nature.

4.3 Biological closure, predicted means and failed responses

Thirty size-selected *Dunaliella* lineages were regrown separately in the same medium (Malerba et al., 2018). Large-selected and small-selected treatment means differ by a factor of 10.4 in cell volume, while their geometric mean total biovolumes differ by only 2.5%. Control geometric mean capacity is approximately 18–21% higher than the two extremes. Thus the close extreme-treatment comparison does not establish uniform capacity across all lineages or treatments.

Frozen equal-biovolume forecasts give held-out errors of 0.124 and 0.179 log units, outperforming a fitted size law and equal cell number under the declared comparisons. An assigned carbon-cost exponent is not distinguished from equal biovolume. The combined-data cost degree 1.02 is a post-hoc description. This is approximate budget closure across separate cultures, not allocation among coexisting size classes; the limiting resource was not directly assayed (full comparisons).

The cell-cycle mechanism transfers a fitted lineage distribution to separate *E. coli* batch-culture summaries (Gangan and Athale, 2017). Mean errors are 2.72% and 0.447%, versus 20.32% and 7.07% for lineage sampling, but variability is overpredicted by 4.61% and 19.69%; the joint criterion fails. No target parameter is fitted. Rounded summaries, length as a mass proxy and unverified physiological equivalence limit the result. Supporting material retains the distribution formulas and full comparison.

A nutrient-pulse experiment provides a direct failed response forecast. In twelve held-out chemostat polycultures, the frozen two-budget C:N reweighting rule allows at most 0.05–0.12 total variation, whereas observed maxima over days 0–12 were 0.29–0.49. The predicted direction holds in only four vessels, and none of the frozen forecasts improves appreciably on persistence. None met the declared recovery criterion by day 12 (response report; Wojcik et al., 2025).

A retrospective follow-up allows arbitrary nonnegative C:N prices, measured preliminary quota changes and every pooled-species mixture. Even so, observed redistribution exceeds the resulting bound in 17/36 comparisons across eight of twelve vessels. Under transferred quotas and fixed allocation opportunities, price adjustment alone cannot explain the failure. Separate-culture quotas do not establish mixed-vessel physiology, and assay uncertainty is unavailable. This conditional diagnosis identifies what a revised explanation must change: physiology, opportunities or the specified relation. Complete bounds are in supporting section S5.10.

A subsequent independent-assay pilot also fails: one eligible test culture has 34.9 million cells/mL, versus 5.55 million predicted from fixed phosphorus quota and 4.24 million from an independent X-ray quota. Measurement independence does not guarantee compatible inputs (supporting S5.11). A field audit passes all 18 primary carbon/nitrogen partial-stock checks, but incomplete

group coverage and undefined allocation opportunities still prevent an allocation test (supporting S5.12).

4.4 Effective-resource inference and competing forecasts

On a fixed vocabulary, a counting measure over word types and candidate symbolic cost $Q_w = L_w^{\theta_L} P_w^{\theta_P}$ imply $p(w) \propto L_w^{-\theta_L} P_w^{-\theta_P}$. Here L and P are letter and dictionary phoneme counts. They do not measure acoustic energy or meaning. Training on three genres gives exponents 0.5153 and 1.8275. The vocabulary and forecasts are fixed before decoding two withheld genres (linguistic study).

The composite improves held-out log loss over fitted letter power by 0.1843 nats per token; the paired document-bootstrap interval is $[-0.1967, -0.1696]$ for composite minus letter-power loss. Smoothed training frequencies nevertheless perform much better: losses are 6.136 for frequencies, 7.055 for the composite and 7.239 for letter power. Inferred-cost shares also remain far from uniform. The result establishes transferable information in combined symbolic features, while physical resource identification and neutral allocation remain unresolved. Contextual and information-based explanations are relevant comparisons (Piantadosi et al., 2011).

4.5 Geometric predictions and observation limits

Locally registered tests compare consequences fixed before outcome fitting. Aftershock productivity gives a primary exponent 0.63 with 95% interval $[0.51, 0.71]$, excluding both rupture-area and moment predictions, 1 and 1.5. Changing to rupture-length counting windows gives 0.90, $[0.76, 0.97]$. The substantial window dependence qualifies the interpretation. Flare ribbon area gives density exponent 3.163, $[2.962, 3.457]$, excluding predictions 2 and 7/3. Both registrations call these outcomes inconclusive because their rules do not choose either candidate; exclusion of the individual predictions is retained (test ledger).

Global lake radii initially give survival exponent 1.930 and iid equivalence to 2. Replacing the global projection with physical local geometry gives 1.859, with basin-bootstrap interval $[1.785, 1.924]$. Exact 2 is excluded under that analysis, but the interval overlaps the frozen equivalence band $[1.9, 2.1]$. The original positive iid conclusion does not survive this geometry correction. Two local projections agree; topology, completeness and classification still limit a physical explanation (follow-up).

The earthquake b–hypocentre-dimension comparison stops earlier. Southern California provides six sequences passing population and resolution gates, below the required ten. Italian CLASS/HORUS provides many calibrated proxies, but a frozen magnitude-estimator check fails in 45 of 56 required synthetic scenarios, including 30 of 32 noiseless converted-magnitude cases. Conversion can leave gaps that invalidate a uniform-bin correction. Neither attempt produces a real b–dimension hypothesis comparison. These are eligibility and measurement findings, not physical evidence against either relation.

4.6 Independent constraints in chemical and sediment transport

For CO2 and CH4 on reference zeolite RM8850, we calibrated a shared-capacity Langmuir model from pure gases and sealed predictions before evaluating 45 mixture conditions at three temperatures (Wedler et al., 2024). Counting measure over species and vacancies defines the comparison; calibrated binding affinities and reservoir fugacities specify the unequal allocation. Capacity is effective within each temperature. Balanced component MAE is 0.247 mmol/g, versus 0.620 when competition is omitted: a 60% improvement. All temperatures pass the frozen tolerance, defined relative to maximum pure loading. Established IAST gives 0.243 mmol/g, so no

predictive advantage over adsorption theory is demonstrated. Only 39/90 component predictions lie within reported observation uncertainties; calibration-range extrapolations also remain. This supports transferable constraints at finite accuracy, with the retrospective chronology, established mechanisms and complete diagnostics in supporting section S5.14.

A retrospective sediment transfer calibrates 19 narrow-channel sphere runs and predicts nine wider-channel runs using separate grain assays (Deal et al., 2026). Measured shape reduces balanced flux error by 42.6% versus unit shape; the fuller wall correction reduces it by 77.2% versus hydraulic radius. The declared accuracy tolerance is permissive, low-flux errors exceed fourfold, and separate old-material fits predict better. This tests a constitutive constraint transfer under material-balanced counting measure, not neutral log-/linear-orthopolic size allocation. One target was exposed before freezing forecasts; complete comparisons and limitations are in supporting S5.15.

5. Learning from validity, departures and failure

A general law earns scientific content through its domains, consequences and limits. Orthopolity is exact for complete resource hierarchies and is realized by the specified exchange and physical models. Its broader empirical reach requires determining when comparable concentrations receive comparable resource, which effective resource is relevant, and what modifies that allocation.

Three kinds of finding should remain distinguishable. First, a metric or sampling change can alter the apparent distribution without changing the physical system: radial power and flux density, mode energy and spectral radiance, or lineage and population sampling are examples. The correction is justified independently of the desired result. Second, a physical constraint can produce a predicted non-neutral profile: measured absorption, quantum level spacing, birth-cycle crossings and pure-calibrated adsorption affinities supply such explanations. Third, a specified resource model can fail its own forecast. The chemostat reweighting, bacterial variability and cantilever spatial-loss errors remain failures until a revised explanation succeeds on additional evidence.

Observed inequality is therefore not by itself a diagnosis. Conversely, an arbitrary choice of resource, measure or constraint can make any positive profile satisfy an allocation identity. Such a construction does not establish a physical law. An application must state which inputs were independently known, what was learned from calibration, what transfers unchanged and what observation would contradict the specified model.

An unavailable test cannot decide between physical resource models; a qualified forecast that fails should narrow the claimed domain or challenge its assumptions. The next comparisons should change independently specified requirements or constraints so that rival explanations predict different relaxation rates, allocation profiles or cross-environment contrasts such as (4). A success supports the specified relation; a failure can identify a missing interaction or a boundary.

6. Contribution and conclusion

Orthopolity links resource concentration, incidence and allocation as a general law. Finite capacity and complete accounting establish its exact core; specified symmetries and dynamics realize its allocation extension. Known physical laws clarify which equalities hold and how independently specified constraints alter the observable profile. Restricted inverse inference then makes abundance informative through predictions that transfer between environments.

The empirical results establish useful conditional successes, failed forecasts and observation boundaries. They do not yet demonstrate a common neutral regime across the natural populations

examined. This limits the present evidence for universal physical reach while preserving the ambition to establish the law. The strongest further contribution will be a resource explanation that predicts both where the relation holds and how a specified change makes it depart or fail.

Data and code availability

The repository retains source notices, fixed configurations, forecasts, algorithms and outputs. The supporting material contains the moved technical detail, full evidence comparisons and a map to study-specific reports. The run registry distinguishes empirical measurements, retrospective analyses and simulations; earlier failed results are preserved. Source-specific licences and conditions are in data/NOTICE.md.

The make all target verifies the original frozen inputs, runs the tests and regenerates the original analysis suite. Later studies have separate offline reproduction targets documented in their reports. The make paper target builds this article and its supporting PDF; registry-verify checks retained study references. Published FIRAS reconstruction and bacterial figure summaries are identified as such, and neither is presented as a new raw-data experiment. The supplied 2017 manuscript was inspected privately and is not redistributed.

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